Despite the skin's defenses, at times pathogenic microorganisms gain access to the tissue under the skin surface. Here, they encounter a specialized set of cells called the skin associated lymphoid tissue (SALT) (Figure 13.9). The major function of SALT is to confine microbial invaders to the area immediately underlying the epidermis and to prevent them from gaining access to the blood stream. One type of SALT is the langerhans cell, a specialized myeloid cell that can phagocytose antigens.

13.3 Cells of the Immune System

All blood cells arise from a type of cell called the hematopoietic stem cell(HSC). Stem cells are cells that can differentiate into other cell types. They are self renewing and they maintain their population level by cell division. This chapter describes the formation of blood cells and the properties of the various cells of the immune system.

13.3.1 Hematopoiesis

Hematopoiesis is the formation and development of blood cells of all types. In humans, hematopoiesis begins in the yolk sac in the first weeks of embryonic development. As gestation continues, the site of hematopoiesis gradually shifts to the bone marrow such that it becomes the principle site at the time of birth.

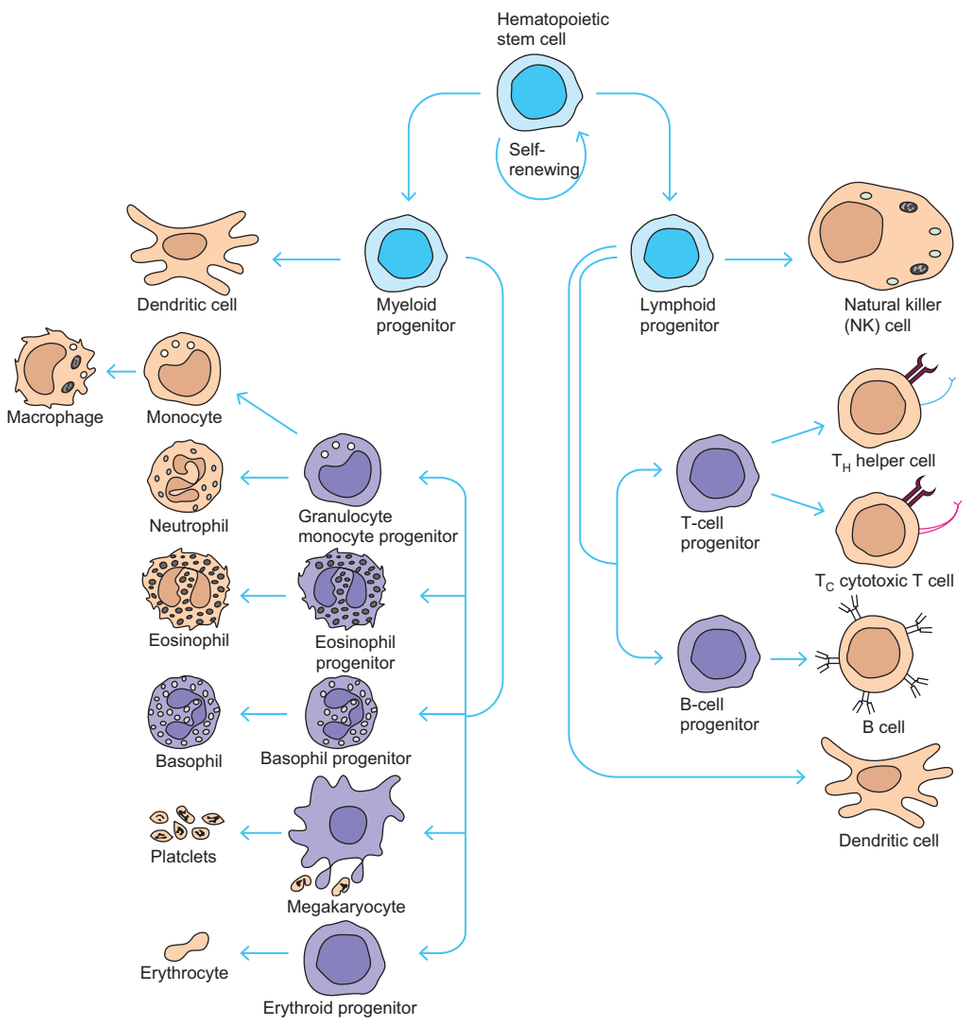


Figure 13.10: Hematopoiesis

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